

The Tiny Object Holding Up the Modern World

A lens into semiconductors and the geopolitics around them.

This is not an academic article filled with professional jargon. This is merely an article that reflects my understanding of a globally important issue. Give it a read if this isn't an area you are familiar with.

For those of us who studied physics in high school, we might be familiar with the words conductors and insulators. In simple words, conductors are materials through which electricity moves easily, while insulators are materials through which electricity does not move easily. But what are semiconductors? And why is this small piece of technology at the centre of almost every activity in our day-to-day lives? And, perhaps more importantly, why are governments willing to spend billions trying to manufacture them?

Semiconductors are materials that allow some electricity to pass through them, which can be manipulated to different conditions allowing them to create electrical switches. These switches form transistors, which can be combined to perform logical operations through logic gates. Logic gates form circuits, and these circuits make up processors and circuits that power smartphones, laptops, cars, medical equipment and countless other devices.

So how are these semiconductors placed in our devices? When transistors are arranged in precise patterns they form circuits, commonly known as chips. These chips are built on extremely pure, thin layers of silicon called wafers. Microchips are built by creating complex patterns and layers on this wafer. How are these patterns mapped onto the silicon wafer? Through photolithography, a process where light carrying a pattern is projected through a mask and optical system onto a photosensitive wafer. So essentially, chip manufacturing is like repeatedly printing precise patterns, layer after layer, until a structure capable of performing complex tasks is created.

But, how do humans physically manufacture billions of these chips?

The manufacturing process of a chip is not a simple, independent process. Companies design chips, companies develop the software used to design them, companies manufacture specialised equipment, companies fabricate the chips, and other companies package and test them before they eventually become part of a product.

Fabless companies, such as NVIDIA, Advanced Micro Devices (AMD), Qualcomm and Apple, are companies that design chips not manufacture them. Then we have foundries, companies that actually manufacture chips designed by other companies. TSMC (Taiwan Semiconductor Manufacturing Company) is the world's leading pure-play foundry, while Samsung also operates a major foundry business. Then we have Advanced Semiconductor Materials Lithography (ASML), a company that makes the machines for lithography. ASML is particularly important because its extreme ultraviolet (EUV) lithography technology is unique to the company and is used in the production of the most advanced chips.

So, to simplify the big cycle: a company such as NVIDIA designs a chip, companies such as ASML provide some of the equipment needed to manufacture it, and a foundry such as TSMC manufactures the chip. This means that developing a modern semiconductor is not a one-company system. It is a global system built on dependencies.

Silicon is the primary material used for most semiconductor chips. Alternative materials such as gallium and germanium enable specialised applications, but the vast majority of modern integrated circuits are still built on silicon. Silicon itself is not particularly scarce, it makes up more than 25% of Earth's crust and China is the largest producer of silicon metal. The United States has important high-purity quartz resources, which are used in making the equipment/materials needed to produce semiconductor grade silicon.

All this to say that semiconductors are important, which means companies care about them. And these systems are designed with dependencies which means control over critical stages of development creates an influence over who dominates the system.

The geopolitical importance of semiconductors was significantly highlighted during the pandemic when trade routes were compromised and nations experienced shortage of them across industries. Different parts of the semiconductor chain are geographically concentrated. Taiwan's TSMC, the power position in this play, manufactures many of the world's most advanced logic chips and has moved into 2nm mass production. Right now, Taiwan is considered to be the dominant because 90% of leading-edge logic chips are produced there. The US has the world's most influential fabless chip designers, NVIDIA, producing chip designs for GPUs and AI accelerators. Its system accelerates AI Data Centers however, it is heavily dependent on Taiwan's manufacturing company. Netherlands has ASML which is the largest semiconductor lithography equipment manufacturer. Its position is highly exposed to geopolitics and export controls because governments regulate where its advanced equipment can be sold. South Korea positions itself as another strong player in this industry with Samsung operating across memory, foundry and system semiconductors. It forms a direct competition to TSMC. The US also includes Intel and other applied material companies who are designing chips and are trying to become a major external manufacturer. Interestingly, Due to the US-China conflict, the US restricted access to manufacturing material to China. China is now investing in SMIC (Semiconductor Manufacturing International Corporation) for creating domestic production and manufacturing of semiconductors.

So what does this mean for the future?

I think the future of semiconductors is going to be less about who can make the most chips and more about who can reduce their dependence on everyone else. Countries are already investing heavily in domestic manufacturing, critical minerals, research and technology. I think we will see countries trying to become less dependent on others by building their own strengths and working more closely with trusted countries. The competition will be about AI, energy, minerals, equipment, skilled people and the infrastructure. And that makes the semiconductor

industry a fascinating example of how technology, business and geopolitics are becoming increasingly difficult to separate.

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